

PHYS0525 Analog and Digital Electronics – Spring 2026

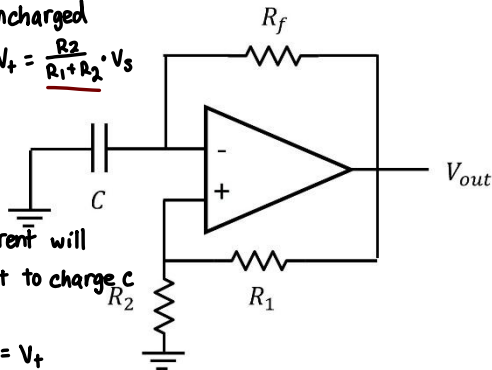
Assignment 9

Due Monday, April 13, 2026

1) Compute the oscillation period for the relaxation oscillator circuit shown below.

Assuming Starting Point

- $V_{out} = +V_s$ + C uncharged
- $V_{in}^- = 0 < V_{in}^+ = V_t = \frac{R_2}{R_1 + R_2} \cdot V_s$



- Since $V_{out} = V_s$ a current will flow through R_f + start to charge C with $\tau = R_f C$
- eventually $V_{in}^- > V_{in}^+ = V_t$
- then $V_{out} \rightarrow -V_s$
- Cap starts to discharge
- then output flips back

$$\text{at } t=0: V_c(t) = V_s + (V_c(0) - V_s)e^{-t/\tau}$$

$$\text{and sub- } V_c(0) = -\beta V_s, V_c(t) = \beta V_s$$

$$\text{thus: } \frac{\beta V_s}{V_s} = \frac{V_s + (-\beta V_s - V_s)}{V_s} e^{-t/\tau}$$

$$\hookrightarrow \beta = 1 - (-\beta - 1)e^{-t/\tau} \Rightarrow \beta = 1 - (1 + \beta)e^{-t/\tau}$$

$$\hookrightarrow \beta - 1 = -(1 + \beta)e^{-t/\tau} \Rightarrow 1 - \beta = (1 + \beta)e^{-t/\tau}$$

$$\Rightarrow e^{-t/\tau} = \frac{1 - \beta}{1 + \beta}$$

$$\hookrightarrow t/\tau = \ln \left[\frac{1 + \beta}{1 - \beta} \right]$$

$$\hookrightarrow t = \tau \ln \left[\frac{1 + \beta}{1 - \beta} \right] \left(\frac{1}{2} \text{ period} \right)$$

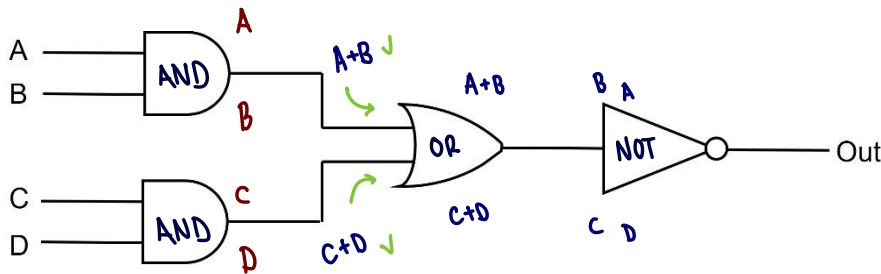
$$T = 2\tau \ln \left[\frac{1 + \beta}{1 - \beta} \right] \text{ and w/ } \beta = \frac{R_2}{R_1 + R_2}$$

$$\hookrightarrow T = 2R_f \cdot C \ln \left[\frac{1 + \frac{R_2}{R_1 + R_2}}{1 - \frac{R_2}{R_1 + R_2}} \right]$$

we resolve

$$T = 2R_f \cdot C \ln \left(\frac{R_1 + 2R_2}{R_1} \right)$$

2) Construct the truth table for the AND-OR-INVERT logic gate shown below. The circuit has four inputs A,B,C, and D and one output. (I.e. write out a table that gives the output of the circuit for all combinations of 0 or 1 input values on each of the inputs.)



Input				Output
A	B	C	D	
0	0	0	0	1
1	0	0	0	1
1	1	0	0	0
1	0	1	0	1
1	1	1	0	0
0	1	0	1	1
0	0	1	1	1
0	1	1	1	0
1	0	1	1	0
1	1	0	1	0
1	1	1	1	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0